(cat, dog, tiger, human). Eurythermy can be an evolutionary advantage: adaptations to cold temperatures (cold-eurythemy) are seen as essential for the survival of species during ice ages. In addition, the ability to survive in a wide range of temperatures increases a species' ability to inhabit other areas, an advantage for natural selection. Eurythermy is an aspect of thermoregulation in organisms.

Those organisms which can tolerate only a narrow range of temperature are **Stenotherms** (Fish, Frogs, Lizards and Snakes).

Over the course of time, by evolution, animals of different ecological habitats have developed different variations and adaptations to temperature changes. It enabled them to survive in different habitats and develop niches. In case of extreme temperatures, organisms have adapted by forming heat resistant spores, cysts (*Entamoeba*), antifreeze proteins (Arctic fishes). Hibernation (winter sleep) and Aestivation (Summer sleep) are useful adaptations to overcome extreme winters and summers. In certain conditions, migration is an appropriate adaptation to overcome extreme temperatures and resultant water and food scarcity. **(Fig 11.2).**

Light

It is an important and essential abiotic factor. Ecologically, the quality (wavelength or colour), the intensity (actual energy in gram calories) and duration (length of day) of light are considered significant for organisms.

Light influences growth, pigmentation, migration and reproduction. The intensity and frequency of light influences metabolic activity, induce gene mutations (UV, X- rays). Light is essential for vision. This is proved by the poorly developed or absence of eyes in cave dwelling organisms. Diapause is also influenced by light in animals. Gonads of birds become more active with increasing light in summer. Light influences the locomotion and movement of lower animals.

Phototaxis: The movement of organism in response to light, either towards the source of light as in Moths (positive phototaxis) or away from light (Euglena, Volvox, earthworm (negative phototaxis).

Phototropism: The growth or orientation of an organism in response to light, either towards the source of light (positive phototropism) as seen in Sunflower, or a way from light (negative phototropism) as in case of the root of plants.

Photokinesis: A change in the speed of locomotion (or frequency of turning) in a motile organism or cell which is made in response to a change in light intensity is called Photokinesis. It involves undirected random movement in response to light.

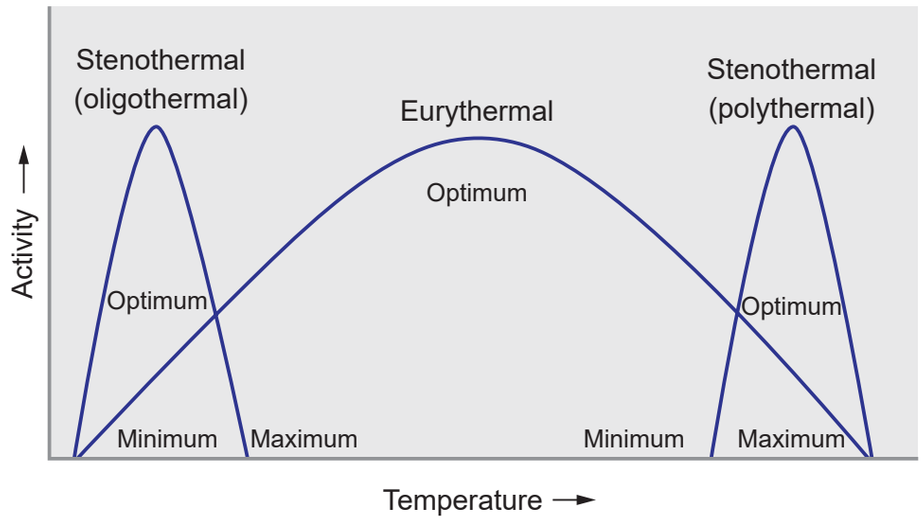


Fig. 11.2 Types of Organisms based on temperature tolerance

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